

Appendix A v1.4.1

Author Errata and Verification Patch

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Status: FROZEN AUTHOR ERRATA / DOCUMENTATION-AND-TOOLING PATCH

Parent frozen release: Appendix A v1.4 - OSF U7M6H - DOI 10.17605/OSF.IO/U7M6H

Purpose: This document records corrections and verification-tooling clarifications to the frozen Appendix A v1.4 release. It does not change the theorem, the exact coefficient certificate, the symbolic derivation, the parent phenomenological model, or any numerical result.

Author statement

This v1.4.1 patch is authored and issued by Prince Upadhyay, Independent Research. The historical Appendix A v1.4 registration remains unchanged and is not retrospectively rewritten.

1. Theorem status

The v1.4 theorem remains unchanged. For strictly positive reduced parameters α , μ , g , β , δ , K and C on the exact e-driven Hopf surface

$$e_c = (\alpha+g)(\alpha+\mu)(g+\mu) / [\beta \delta (\alpha+\mu+g)]$$
$$\omega_c^2 = \alpha \mu g / (\alpha+\mu+g)$$

the first Lyapunov coefficient satisfies

$$l_1 > 0$$

using the convention

$$l_1 = \text{Re}(G_{21}) / (2 \omega), \text{ with } \langle p, q \rangle = 1.$$

Together with positive transversality as e increases, this gives a subcritical Hopf bifurcation throughout the covered interior e-driven family. Locally, the small-amplitude periodic orbit is unstable and lies on the $e < e_c$ side, while the equilibrium loses stability as e crosses e_c upward.

2. Exact positivity certificate

After imposing the Hopf identity $\omega^2 = \alpha \mu g / (\alpha + \mu + g)$, the symbolic reduction gives $\text{Re}(G_{21}) = N/D$. The exact numerator N contains 93 monomials and the denominator D contains 28 monomials. Every exact coefficient in both polynomials is strictly positive.

The unchanged certificate is `exact_positive_polynomial_certificate_v1_4.json`.

Certificate SHA-256:

34439e86255d7d9e61c483e3c97044e247d9eae09bc2c00b13791704c8a68667

3. Verification logic

The coefficient verifier checks all 93 numerator coefficients and all 28 denominator coefficients. That check establishes $N/D > 0$ on the positive orthant. By itself, it does not establish the identity $N/D = \text{Re}(G_{21})$.

The analytic identity comes from the symbolic derivation preserved in the frozen v1.4 release. The v1.4.1 verifier additionally provides an optional independent numerical identity check: N/D is compared with $\text{Re}(G_{21})$ computed from numerical eigenvectors at 50-digit precision over 25 deterministic random positive parameter points.

Recorded identity check: worst relative difference = $1.53e-47$.

4. Transversality

The characteristic polynomial is

$$\lambda^3 + (\alpha + \mu + g) \lambda^2 + (\alpha \mu + \alpha g + \mu g - \beta \delta e) \lambda + \alpha \mu g.$$

At the Hopf point, the crossing derivative is

$$d \text{Re}(\lambda) / de = \beta \delta (\alpha + \mu + g)^2 / (2 D_3) > 0,$$

where D_3 is the positive-coefficient cubic factor appearing in the exact denominator. Hence the eigenvalue pair crosses with positive transversality as e increases.

5. Corrections recorded in v1.4.1

1. The stale candidate-status wording in the historical proof-audit artifact is corrected in the v1.4.1 audit.
2. The verifier's default certificate filename now matches the shipped certificate.

3. The verifier's logical scope is stated explicitly: coefficient positivity proves $N/D > 0$, not the identity $N/D = \text{Re}(G21)$ by itself.
4. An optional independent numerical identity check is included.
5. The verifier output is regenerated without the unrelated environment traceback present in the historical v1.4 artifact.
6. The transversality calculation is recorded explicitly.
7. The first-Lyapunov convention and the local bifurcation side are stated explicitly.

6. Reproducibility dependency

The analytic identity $\text{Re}(G21)=N/D$ originates in the symbolic derivation script `hopf_ll_symbolic_positivity_v1_4.py`, preserved in the frozen Appendix A v1.4 archival release (OSF U7M6H; DOI 10.17605/OSF.IO/U7M6H). Appendix A v1.4.1 is an errata and verification-tooling patch and should be read together with that frozen parent release for full symbolic reproduction.

The v1.4.1 bundle contains two verification-output files with duplicate successful identity-check content. This redundancy has no effect on the theorem or verification result and is retained as part of the frozen patch record.

7. Claim boundary

This document records a mathematical result and its verification within the specified phenomenological dynamical family. It does not establish a microscopic law of nature, does not derive the observer-entropy to horizon-entropy bridge, does not resolve Gate 0, does not establish a global basin-boundary theorem, and does not by itself establish that nature realizes the model.

8. Provenance

The frozen Appendix A v1.4 / U7M6H record is not replaced, rewritten, or retrospectively modified. v1.4.1 is an additive author-issued errata/documentation-and-tooling patch.

9. v1.4.1 bundle inventory

File	SHA-256
CHANGES_v1_4_1.md	c70e4be805b504ca8d0d640c4cbb1238bf406c3d76ec3a855f5d1aa92568e166
PROOF_AUDIT_v1_4_1.md	eaca49cfada8c9b4f2da83bd8e192d4f9e0144d8712b5bc7fb0d9f79984a4535
README_v1.4.1.md	26ebb6f4436ea60e5fc515e81a27dfa367e5acb416f3660073948b04bb67a594
exact_positive_polynomial_certificate_v1_4.json	34439e86255d7d9e61c483e3c97044e247d9eae09bc2c00b13791704c8a68667
manifest_v1_4_1.json	0e13b9aab8f549145124dfe77b13f0975dae59d411b579cfe0286c4f79e2a41f
verifier_run_coefficient_only.txt	363c0bbddc8c8fee00480364324a1e164d7131376936125fd4d00507e9baea8c
verifier_run_with_identity.txt	35a429c9455a53b50ee4217b242f544db42a0126c749c0cee82954d32fa957cc
verify_positive_certificate_output_v1_4_1.txt	35a429c9455a53b50ee4217b242f544db42a0126c749c0cee82954d32fa957cc
verify_positive_certificate_v1_4_1.py	01e70c2584289149636da8087443b1819cf69e9b2b2dce81af653a7ec83de443

End of Appendix A v1.4.1 - Author Errata and Verification Patch